SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

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CALIBRATION

Valid To: April 30, 2018 Certificate Number: 1876.02

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Dimensional

Parameter/Equipment	Range	CMC[2, 5] (±)	Comments
Calipers, Verniers[3]	Up to 36 in	(43 + 2.4*L*) µin	Gage blocks and rod standards
Height Gages[3]	Up to 36 in	(33 + 2.7*L*) µin	Gage blocks and rod standards
Micrometers[3] –			
Outside	Up to 36 in	(28 + 2.6*L*) µin	Supermicrometer™, gage blocks and comparator.
Inside	Up to 36 in	(28 + 2.6*L*) µin	
Depth	Up to 36 in	(28 + 2.6*L*) µin	
Dial and Test Indicators[3]	Up to 2 in	(58 + 0.4*L*) µin	Gage blocks/dial gage tester

Parameter/Equipment	Range	CMC[2, 5] (±)	Comments
Plug/Pin Gages	Up to 6 in	$(23 + 5L)$ µin	Supermicrometer™, gage blocks, and comparator
Verification of Length[3] – Rules and Tapes	Up to 360 in Up to 1200 in	$(0.058 + 9.3 \times 10^{-6}L)$ in $(0.039 + 3.1 \times 10^{-5}L)$ in	Gage blocks, linear standards
Gage Blocks	(0.3125 to 4) in	$(2.6 + 1.3L)$ µin	Gage block comparator
Long Gage Blocks	(5 to 20) in	$(1.4 + 2.6L)$ µin	Comparison of gage blocks using Brown & Sharpe digital probe with amplifier
Thread Measuring Wires	(4 to 120) TPI	$(21 + 6.8L)$ µin	Gage blocks and Supermicrometer™ TPI is threads per inch
Length Standards – Setting Rods	 Up to 40 in	 $(1.4 + 2.6L)$ µin	 Gage blocks, Brown & Sharpe digital probe with amplifier
Thread Plugs – Standard 60° Screw, Metric Screw Major Diameter Pitch Diameter	 Up to 6 in Up to 6 in	 $(18 + 3.2L)$ µin 170 µin	 Supermicrometer™ with thread measuring wires
Thread Plugs – Pipe Inch (NPT, NPSM, NPSL, ANPT, NPTF) Major Diameter Pitch Diameter	 Up to 6 in Up to 6 in	 $(20 + 3.3L)$ µin 180 µin	 Supermicrometer™ with thread measuring wires

Parameter/Equipment	Range	CMC [2, 5] (±)	Comments
Thread Rings – Standard 60° Screw, Metric Screw Pipe Inch (NPT, NPSM, NPSL, ANPT, NPTF) GO / NO-GO	Up to 3 in	210 µin	Master threaded set plug
Ring Gages	(0.001 to 10) in	(18 + 1.5*L*) µin	Sheffield and gage blocks/master rings
Surface Plates[3] – Repeatability Only	(18 in × 24 in) to (72 in × 96 in)	50 µin	Repeat-o-meter

II. Mechanical

Parameter/Equipment	Range	CMC[2, 5] (±)	Comments
Balances[3, 4]	0.01 mg to 1 g (1 to 10) g (10 to 100) g (100 to 500) g (500 to 1000) g (1 to 10) kg (10 to 18) kg (18 to 50) kg (50 to 100) kg (100 to 150) kg	0.036 mg + 0.6*R* 0.051 mg + 0.6*R* 0.25 mg + 0.6*R* 1.2 mg + 0.6*R* 0.02 g + 0.6*R* 0.23 g + 0.6*R* 0.26 g + 0.6*R* 13 g + 0.6*R* 13 g + 0.6*R* 16 g + 0.6*R*	Verification with Class 4, & F weights
Scales[3, 4]	Up to 10 000 lb	0.017 % of reading + 0.6*R*	Verification with Class 6 and Class F weights per NIST HB44
Pressure Gages, Transducers and Transmitters[3]	(-15 to 300) psi (50 to 10 000) psi	0.084 psi 0.2 % of reading	ASME B40, Druck 610 Deadweight tester

Parameter/Equipment	Range	CMC[2] (±)	Comments
Verification of Force[3] – Tension	 (0 to 1000) lbf	 0.01 % of reading	 Class F weights
Torque[3]	(4 to 50) in·lbf (30 to 400) in·lbf (80 to 100) in·lbf (20 to 250) ft·lbf (100 to 1000) ft·lbf	0.15 in·lbf 1.2 in·lbf 2.9 in·lbf 0.73 ft·lbf 3.1 ft·lbf	Torque transducers and indicators

III. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature – Measure[3]	(0 to 260) ºC	0.49 °C	Fluke 725 with Pt 100 Ω RTD
Temperature – Measuring Equipment[3] Ice Point	(Ambient + 5 to 100) °C (100 to 300) °C 0 °C	0.61°C 1.0 °C 0.57 °C	Hart Scientific 9100 dry block ASTM E563-02
RTD – Simulation[3] Pt 385, 100 Ω Pt 385, 1000 Ω	 (-200 to 800) °C (-200 to 630) °C	 0.49 °C 0.39 °C	 Fluke 725

Parameter/Equipment	Range	CMC[2] (±)	Comments
Thermocouple Simulation[3] –			
Type K	(-200 to 0) °C (0 to 1370) °C	1.6 °C 1.2 °C	Fluke 725
Type J	(-200 to 0) °C (0 to 1200) °C	1.4 °C 1.1 °C	
Type T	(-200 to 0) °C (0 to 400) °C	1.6 °C 1.2 °C	
Type E	(-200 to 0) °C (0 to 950) °C	1.3 °C 1.1 °C	
Type R	(-20 to 0) °C (0 to 500) °C (500 to 1750) °C	3.0 °C 2.2 °C 1.8 °C	
Type S	(-20 to 0) °C (0 to 500) °C (500 to 1750) °C	3.0 °C 2.2 °C 1.8 °C	
Relative Humidity – Measure[3]	(0 to 80) % RH	3 % RH	Vaisala HMI41 w/ HMP42
Ovens, Furnaces, and Environmental Chamber Uniformity[3]	(Ambient +20 to 500) °C	1.3 °C	ASTM E145-01 (temperature only)
Infrared Temperature Measuring Equipment[3]	Ambient to 100 °C (100 to 500) °C	1.3 °C 2.0 °C	Hart Scientific 9132 black body

IV. Time & Frequency

Parameter/Equipment	Range	CMC[2, 5] (±)	Comments
Digital/Mechanical Tachometer[3]	(20 to 29 999) RPM	0.1 % of reading + *LSD*	Direct reflective pickup tachometer

Parameter/Equipment	Range	CMC[2, 5] (±)	Comments
Timers[3]	15 s to 24 hr	The greater of 0.05 % full scale (fs) or 1 *LSD*	Reference stopwatch
Stopwatches[3]	15 s to 24 hr	0.01 % full scale + 0.5 s	NIST synchronized computer clock

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[1] This laboratory offers commercial calibration services and field calibration services (where noted).

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] Field calibration service is available for this calibration and this laboratory meets A2LA *R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

[4] The uncertainty of scale verification is highly dependent on local conditions such as the resolution of the scale. Any statement of CMC would therefore be misleading. The class of the best weights used by the laboratory is shown in the Comments column.

[5] In the statement of CMC, *L* is the numerical value of the nominal length of the device measured in inches, and *LSD* is least significant digit, and *R* is the resolution of the unit under test.

Accredited Laboratory

A2LA has accredited

ALABAMA SCALE & INSTRUMENT - MARSHALL, INC.
DBA ASI CALIBRATION LABS - TEXAS

Marshall, TX

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANIS/NCSL Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

Presented this 22nd day of April, 2016.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 1876.02
Valid to April 30, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.